1.introduction

Given that cancer is one of the major causes of death around the world, and with drug resistance and disease recurrence limiting traditional anticancer treatments like chemotherapy, the need for a novel treatment became increasingly apparent. This novel treatment is cancer immunotherapy, which is important as immune system evasion is one of the main ways used by tumor cells to metastasize and grow. This treatment aims to augment the patient's immune system and help it recognize and eliminate malignant cells. Tumor cells use immune checkpoints to escape the immune system, thus immune checkpoint inhibition has become one of the most effective strategies against tumors in recent years. One of the drugs developed for this purpose is nivolumab. This article discusses nivolumab's pharmacological features, mechanism of action, current clinical applications, combination treatment strategies, adverse effects, and future directions in cancer immunotherapy.

2. Immune Checkpoint Background

The pathogenesis of cancer involves many steps leading to uncontrolled cell division and tumor formation. This tumor can further invade other organs and metastasize by escaping immune detection by the body's immune system. Malignant cells do that by the interaction of the programmed cell death ligand (PD-L1) expressed on them with the programmed cell death protein 1 (PD-1) receptor expressed on T-cells. This interaction suppresses T-cells and leaves them unable to properly kill cancer cells. That's why reactivation of the immune response by inhibitors of this interaction is important for cancer treatment. Moreover, these inhibitors can be used alone or in combination with chemotherapy [1] .

Another class of immune checkpoint inhibitors acts on cytotoxic T- lymphocyte-associated protein 4 (CTLA-4), like ipilimumab. In order for T cells to be activated, 2 interactions must occur. The first one is the interaction of the T cell receptor with antigens presented by the MHC molecule. The second is the interaction of B7 on antigen presenting cells with CD28 present on T cells. Once this activation happens, a molecule called CLTA-4 is upregulated on T cells' surface, and it binds to B7 with greater affinity than CD28. This sends inhibitory signals to T cells, suppressing cytotoxic T cell function. Consequently, CLTA-4 inhibitors have been developed as an immunotherapy to block this interaction, resulting in sustained T cell activity and potent anti-tumor responses [6].

3.Nivolumab: characteristics and pharmacology

Nivolumab is an IgG4 immune checkpoint inhibitor antibody that binds to (PD-1) preventing PD-L1, PD-L2 interaction. Restoring T cell activity by blocking (PD-1) mediated inhibitory signaling. Therefore, restoring the antitumor response of the immune cells [2].

Nivolumab is given intravenously, and has a half-life of approximately 26–27 days. Like other protein drugs, it is metabolized by proteolytic enzymes and degraded into amino acids. So, it doesn't undergo hepatic metabolism through cytochrome enzymes ,minimizing the potential of drug -drug interactions [3].

4.Nivolumab: Clinical Applications and Combination Therapy

Nivolumab is effective in treating different types of advanced cancers, especially those resistant to standard treatments. It improves overall survival and is more tolerated than chemotherapy overall. It's used in the treatment of advanced non-small cell lung cancer (NSCLC), advanced melanoma, advanced renal cell carcinoma ,and many more [4].

Despite improving cancer treatment, almost fifty percent of patients do not reach a good clinical outcome with just a single immune checkpoint inhibitor, which led to the development of combination strategies [6] .The combination of both nivolumab and ipilimumab targets two different immune checkpoints that inhibit T-cell activation. So, by simultaneously blocking these inhibitory pathways, the combination produces a stronger antitumor immune response than either drug alone [6].

The combination improved response rates, progression-free survival, and overall survival in several malignancies as proven by clinical studies. However, a higher incidence of immune-related adverse effects was associated with the combination therapy [6].

5.Nivolumab: Adverse Effects and Resistance

Nivolumab has shown a relatively acceptable safety profile, as demonstrated by a systematic review and meta-analysis of 21 clinical trials including 6173 patients. Importantly, in comparison to conventional treatments such as chemotherapy, nivolumab showed no significant increase in the risk of either serious or fatal adverse effects. But it still can cause serious immune related adverse events (SAEs) and rarely, fatal adverse events (FAEs). The most common serious adverse effects of this drug are pneumonitis, interstitial lung disease, and colitis. Pneumonitis is the main cause of both serious and fatal toxicity, making it the most clinically significant adverse effect. Other serious adverse effects may involve the kidneys, heart, brain, endocrine organs, liver, and many more [7].Although it's rare, nivolumab can also cause severe infusion reactions [5]. Moreover, fatigue, decreased appetite, diarrhea, and nausea are common adverse effects [2].

Despite showing great results in cancer treatment, resistance against immune checkpoint inhibitors is still a problem. Several factors related to tumors such as rapid tumor growth kinetics and tumor heterogeneity contribute to resistance. Furthermore, there are no proven biomarkers that can accurately identify patients who will benefit from PD-1 inhibitors. That's why authors highlight the importance of combination therapy as well as the development of novel biomarkers to enhance patient selection [8].

6.Conclusion

To conclude, nivolumab has revolutionized cancer immunotherapy. However, its clinical use remains limited due to its high cost, intravenous administration, poor tumor penetration, and immune related toxicity. That's why future research aims to develop orally active small molecules of PD-1/PD-L1 inhibitors. Developments in structural biology and screening technologies are expected to speed up the development new PD-1/PD-l1 inhibitors [9].

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